

5(a)	<i>five from:</i> A 'cells' for 'lymphocytes' throughout 1 ref. to antigen presentation ; A macrophage presents antigen 2 recognition / binding (in context of B-, or T-lymphocytes) ; e.g. cell surface receptor <u>specific</u> to toxoid / clonal selection A immunoglobulin / antibody on surface of B-lymphocytes 3 T-lymphocytes or B-lymphocytes, divide by mitosis / clonal expansion ; 4 (some) plasma cells, formed / AW ; R if in context of T-lymphocytes 5 antibody molecules / antibodies, secreted / produced / released ; R if in context of T-lymphocytes 6 T helper cells secrete, cytokines / interleukins ; 7 ref. to action of cytokines / interleukins ; e.g. to stimulate, humoral response / B-lymphocytes / to stimulate <u>macrophages</u> / angry <u>macrophages</u> 8 formation of memory cells ; <i>only in context of B- / T-lymphocyte activation</i>	5
------	--	----------

Question	Answer	Marks
5(b)(i)	<i>three from:</i> 1 failure to distinguish between self and non-self / AW ; 2 specific B-,lymphocytes / cells, are not destroyed (during development) ; 3 antibodies are produced against, cell surface / ACh, receptors on, muscle cell / neurones / nerve cells ; 4 antibodies, bind to / block, receptors ; A antibody–antigen, binding / complex <i>if receptors already given</i> 5 AVP ; any detail related to the disease e.g. (chemical) transmitter molecules / acetylcholine, cannot bind (to receptor) impulses are not transmitted across synapse / neurones cannot send impulses ref. to causing, muscle weakness / muscle fatigue ref. to neuromuscular junction does not function	3
5(b)(ii)	<i>two from:</i> so immune response, occurs all the time / recurs ; (defective) B-lymphocytes still being produced / AW ; I T (helper) cells A not destroyed / remain plasma cells / B-lymphocytes, continue to release antibodies against receptors ; memory cells / immunological memory, present ; receptors are always present (on neurones / nerve cells) ; AVP ; e.g. no cure	2

Question	Answer	Marks
----------	--------	-------